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Spectral properties of Er^{3+} doped oxyfluoride tellurite glasses

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Abstract

The glass forming capability, glass structural organization and optical properties have been examined in rare-earth doped oxyfluoride zinc tellurite glass system. As a function of composition, differential thermal analysis, vibrational spectra, optical absorption, spontaneous emission and lifetime measurements have been analyzed in term of fluorine influence. The bulk composition variation and addition of fluoride compounds in tellurite glasses result among others, broad emission spectra compared to silicate glasses, in improved emission lifetime and slight difference in relative band intensities compared to pure oxide glass.

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1. Introduction

The Wavelength Division Multiplexing in telecommunication systems requires amplifier materials having the broadest possible emission spectrum and an increasing demand is perceived for compact optical amplifiers in order to supply low-cost optical devices. Among the latest evolutions of telecommunication network certainly the research

field concerning the behavior of erbium as a rare-earth metal dopant in glasses is very important [1]. In particular, host glasses based on tellurium oxide doped by Er^{3+} ions can provide a broadband amplification around $1.5 \mu\text{m}$ which corresponds to the window of minimum absorption loss in the silica-based optical fibers [2]. In certain systems as oxyfluoride host glasses, the modification of the rare-earth spectral properties by addition of fluorine might be promising to obtain a broad amplifier bandwidth with an intrinsically flatness gain required in the development of optical transmission network [3].

Moreover, the oxyfluoride glasses are expected to have chemical durability and thermal stability representing a compromise between pure fluoride

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and oxide glass with flexible optical properties. In this context, we were interested in studying the influence on optical behavior (optical absorption, emission properties, up-conversion luminescence and lifetime) of adding increased amounts of fluorine to the Er^{3+} doped zinc tellurite glasses and in parallel, investigating the induced modifications of the host glass network from Raman and IR spectroscopy studies.

2. Experimental

The glass samples (6 g batch) were synthesized from commercial reagent grade powders (TeO_2 , ZnO , ZnF_2 , Na_2CO_3 , Er_2O_3 : Rare Metallic Co.). Doped oxide and oxyfluoride tellurite glasses were obtained by addition to the batch of the desired rare-earth oxide. The mixed powders were melted, in platinum crucible covered with a platinum cap to avoid an important vaporization loss, at 650–800 °C for 20 min; then quenched and annealed for 3 h at 30 °C below their glass transition temperatures. The compositions of glasses were analyzed by AES–ICP (atomic emission spectroscopy–inductively coupled plasma) and by XPS (X-ray photoelectron spectroscopy). The thermal properties have accurately been analyzed by a differential thermal analysis apparatus (Mac Instruments Inc.) at a heating rate of 10 °C min^{-1} in the temperature range of 25–700 °C. The densities were measured using calibrated CCl_4 as liquid medium. Raman scattering spectra were obtained using a spectrometer (Perkin–Elmer instrument, $^{\text{TM}}$ GXR-AMAN FT-IR) with 1064 nm emission line of a

diode pumped Nd^{3+} : YAG laser, 200 mW. The absorption spectra were measured at room temperature using an UV–visible–near infrared (Hitachi U-3500 model) spectrophotometer over a spectral range of 250–3200 nm. IR transmittance spectra were recorded using a Fourier transform infrared (FTIR) spectrophotometer (Perkin–Elmer instrument). The linear refractive indices were determined by the Brewster angle reflection method at 1064 nm (Nd^{3+} :YAG laser). Emission spectra for the 1.5 μm band of Er^{3+} were measured with a 974 nm pumping from a continue tunable near-IR Ti–sapphire laser, also used for the up-conversion excitation (InGaAs detector (900–1600 nm) or Si CCD (350–1050 nm)). Experimental decay rates of $^4\text{I}_{13/2}$ and $^4\text{I}_{11/2}$ erbium levels were measured using laser diode emitting at 975 nm. Emission was collected at 90°, separated from the incident beam by a monochromator (10 nm resolution) and focused on a high speed cooled germanium detector.

3. Results

The analyzed composition of obtained oxyfluoride and oxide zinc tellurite glasses are closed to that of the initial batches (Table 1). Fluorine loss is about 1–4 wt%. Obtained samples appeared to be of good optical quality, with no visual evidence of devitrification. The glass transition temperature observable on the DTA curves follows a continuous tendency of decreasing with gradual introduction of ($\text{ZnO} + \text{ZnF}_2$). The difference $T_x - T_g$ slightly increases with the introduction of fluorine and is large enough (about 20 °C) to obtain stable glasses.

Table 1
Compositions of the studied oxide and oxyfluoride tellurite glasses and the ratio zinc/tellurium and fluorine/oxygen

Glass matrix composition		Composition (mol %)				
		TeO_2	ZnO	Na_2O	ZnF_2	$R_{\text{Zn/Te}}$
ETZN	Oxide	75.0 ± 0.4	20.1 ± 0.1	4.9 ± 0.1		0.27
ETZ	Glasses	75.1 ± 0.4	24.9 ± 0.1			0.33
ETZF-1		74.7 ± 0.4	18.3 ± 0.1		7.1 ± 0.1	0.34
ETZF-2	Oxide	74.6 ± 0.4	8.8 ± 0.1		16.6 ± 0.1	0.34
ETZF-3	Fluoride	69.0 ± 0.3	9.3 ± 0.1		21.7 ± 0.1	0.45
ETZF-4	Glasses	60.5 ± 0.3	12.0 ± 0.1		27.5 ± 0.1	0.65
ETZF-5		46.6 ± 0.2	18.2 ± 0.1		35.2 ± 0.2	1.15

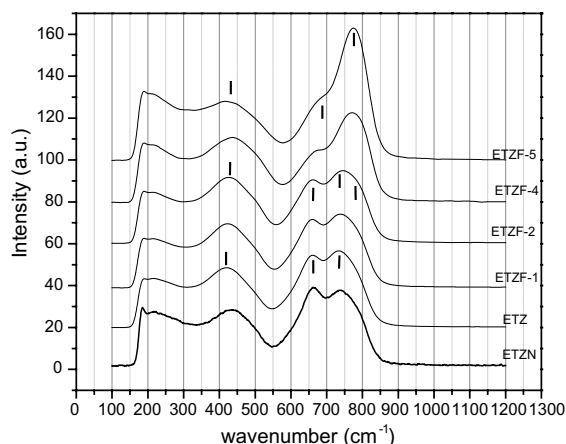


Fig. 1. Raman scattering spectra for a series of studied zinc tellurite oxide and oxyfluoride glasses.

Like oxide glasses with similar composition, the Raman spectra of oxyfluoride glass samples present three main bands around 700–800, 660–670, 410–420 cm^{-1} [4] (Fig. 1). A significant modification of the intensity of the 700–800 and 660–670 cm^{-1} bands is observed with composition variation. A shoulder around 775 cm^{-1} and a slight intensity decline of the band located at 420 cm^{-1} are observed for increasing fluorine concentration.

The tellurite glasses have a large IR transmission spreading up to 6 μm , but the presence of hydroxyl groups is an important problem in the range of middle IR reducing the advantageous characteristics of these glasses. On the spectra represented in Fig. 2, the absorption peaks were normalized with respect to the thickness of the glass samples and a broad absorption band 2500–3600 cm^{-1} and a weaker at 2260 cm^{-1} (4.4 μm) can be distinguished.

The experimental lifetimes values obtained for tellurite glasses ($75\text{TeO}_2\text{--}20\text{ZnO--}5\text{Na}_2\text{O}$ and $75\text{TeO}_2\text{--}20\text{ZnO}$) are around 3, 2 ms and when the proportion of fluorine increases, the lifetime of both $^4\text{I}_{13/2}$ and $^4\text{I}_{11/2}$, significantly increases (Fig. 3). The normalized measured emission spectrum of oxide and oxyfluoride tellurite glass are shown in Fig. 4 illustrating the broad emission spectra around 1.5 μm in these tellurite glasses. The global shape of the peak emission is modified by the different contributions of each components in-

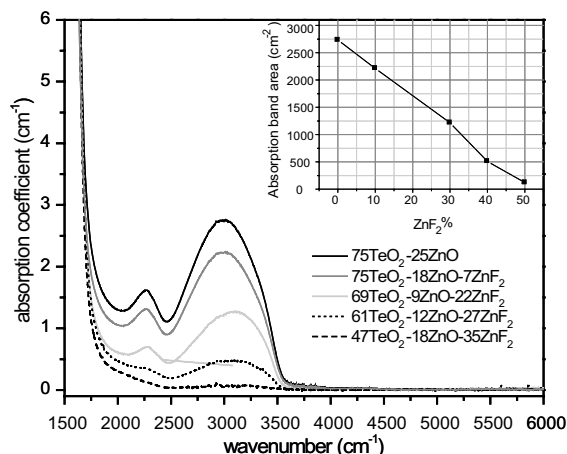


Fig. 2. Infrared spectra for the oxide and oxyfluoride tellurite glasses.

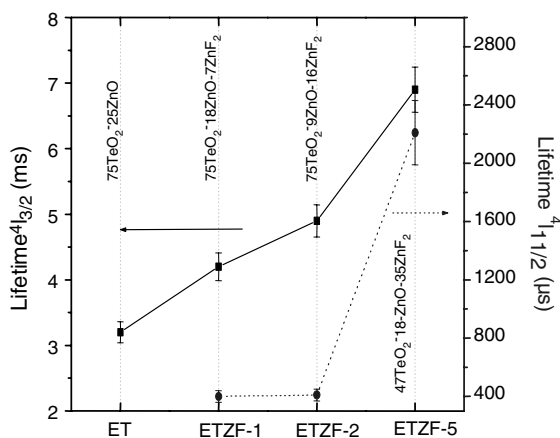


Fig. 3. Evolution of the $^4\text{I}_{11/2}$ and $^4\text{I}_{13/2}$ lifetime through several composition of oxide and oxyfluoride tellurite glasses.

cluded into the broad emission band. In oxyfluoride samples, the shoulder at lower wavelengths beside the maximum of the peak around 1530 nm slightly decreases. The main shoulder around 1560 nm is more intense in relative intensity than those observed on oxide glasses with similar erbium contents. In the case of the fluorine richest glass, the whole profile of emission peak seems to remain unchanged, similar to the other oxyfluoride samples, with the exception of a light narrowing of the emission band supplementary observed. The

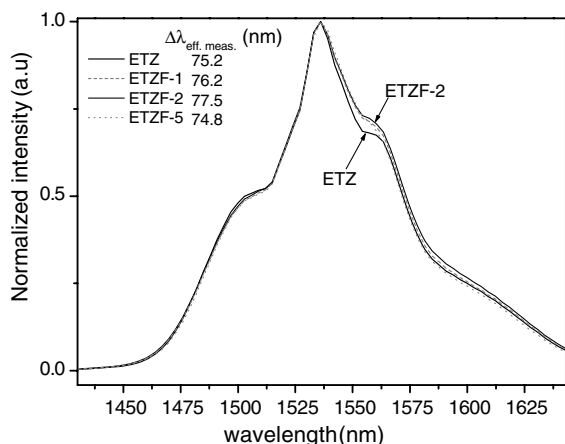


Fig. 4. Normalized emission intensity around 1.5 μm band for oxide and oxyfluoride tellurite glasses and effective emission linewidths ($\Delta\lambda_{\text{eff, meas.}}$).

effective emission linewidths for the normalized measured emission spectra have been calculated and the ETZF-1 and ETZF-2 oxyfluoride tellurite glasses show a wider emission linewidths than oxide tellurite glasses.

4. Discussion

4.1. Properties of the host glasses

At constant ratio $[\text{ZnF}_2 + \text{ZnO}]/[\text{TeO}_2]$ (ETZ, ETZF-1, ETZF-2), a shoulder at high wavenumber (around 775 cm^{-1}) is observed on Raman spectra for growing fluorine concentration and the probable changes due to likely formation of oxyfluoride tellurite entities remain soft. This later vibration, in agreement with Sekiya et al., is attributed to $\text{Te}-\text{O}^-$ non-bridging stretching vibration mode of TeO_{3+1} and TeO_3 groups [4]. In parallel, the slight intensity decrease of the band located at 420 cm^{-1} illustrates the decrease of the $\text{Te}_{\text{eq}}-\text{O}\cdots\text{Te}_{\text{ax}}$ linkages of a continue network made of TeO_n ($n=4, 3+1$ and/or 3) entities. This effect is consistent with the idea of a partial former network break induced by the substitution of some oxygen by F^- atoms; indeed, the fluorine atoms introduced in the oxide network can be considered not to act as a bridge between two Te atoms. Increasing the $[\text{ZnF}_2 + \text{ZnO}]/[\text{TeO}_2]$ ratio results in

a significant modification of the intensity ratio $((I_{740}+I_{775})/I_{660})$ of these bands involving a decrease of the TeO_4 entities and formation of more TeO_{3+1} and/or TeO_3 groups. The obvious intensity increase of the band around 775 cm^{-1} confirms the formation of TeO_{3+1} and TeO_3 with non-bridging oxygen (Table 1).

On transmission spectra, the broad absorption band $2500\text{--}3600\text{ cm}^{-1}$ and the weaker at 2260 cm^{-1} ($4.4\text{ }\mu\text{m}$) can be assigned to the stretching vibration of OH^- groups. The hydroxyl and fluorine ions are isoelectronic, similar ionic size and hydroxyl ions can be easily replaced by fluorine during melting. Therefore, the decreasing of both absorption bands on IR spectra can be interpreted on the basis of an introduced fluorine concentration increase. The damageable quenching effect of OH upon the $^4\text{I}_{13/2}$ levels of Er^{3+} can be consequently reduced.

4.2. Spectroscopic properties of Er^{3+} doped glasses

In the case of tellurite glasses, the observed decay lifetime (3.2 ms) is in the same order of magnitude of the value calculated by Judd–Ofelt method suggesting that the influence of multiphonon relaxation seems to be minor [5]. Indeed, the low-phonon energy typical of tellurite glasses and these oxyfluoride glasses, can favor a quite long emission lifetimes of the $^4\text{I}_{13/2}$ level mostly governed by radiative decay rate. For oxyfluoride tellurite glasses, the experimental lifetime is regularly increased with a fluorine content growth. This trend expected from the estimation of the decay lifetime by Judd–Ofelt method calculation [6] which predicts an increased of experimental lifetime with introduction of fluorine is experimentally confirmed and illustrated by Fig. 3. According to a lower impact of multiphonon loss, the lifetime values recorded for oxyfluoride glasses should be strongly influenced by the presence of fluorine within glass host and additionally, to the increase of the ratio Zn/Te in the ETZF-5 glass case which modify the local electrostatic field strength and symmetry of the ligand field around the rare-earth.

The slight difference of the linewidth and emission peak shape between oxide and oxyfluoride

ride tellurite glasses at 1.5 μm is due mostly to the difference in the local environment of the Er^{3+} ions (Fig. 4). In tellurite glasses, the Er^{3+} ions have a coordination sphere of oxygen, whereas in oxyfluoride glasses they might be surrounded by mixed anions, fluorine and oxygen ions. The decrease of Ω_2 Judd–Ofelt parameters (5.57 and 3.00 for ETZ and ETZF-5 respectively [6]), with increasing fluorine content suggest this increase of the number of fluorine ions surrounding erbium ions. Moreover, the introduction of fluorine for a constant ratio of Zn/Te leads to a slight depolymerization of the glass network presenting several tellurium groups which can also contribute in minor proportion to change the local crystal field. This can explain the increase of the linewidths for ETZF-1 and ETZF-2 resulting in a higher variation in the environment of Er^{3+} from one site to another site. On the other hand, a lesser site-to-site variation can be expected in the case of ETZF-5 where the dimensionality and the connectivity of the glass decrease with increasing the proportion of TeO_3 and TeO_{3+1} groups as it was shown by Raman spectroscopy. The narrowing of the emission peak observed may be related to the homogeneity of the shell-coordination of the erbium ions by increasing the fluorine contents.

5. Conclusion

These glasses have been melted over an extensive compositional region in the range of 75–47 mol% of TeO_2 and present a relatively high chemical durability and a low tendency to devitrification. Addition of fluoride compound results in reduction of Te–O–Te linkages due to a gradual transformation of trigonal bipyramid TeO_4 through TeO_{3+1} to trigonal pyramid TeO_3 . The former network connectivity decrease is strengthened by a possible high ratio $(\text{ZnO–ZnF}_2)/\text{TeO}_2$ compared with pure oxide tellurite glasses. For a constant ratio of Te/Zn, the effective bandwidth of absorption cross-section and normalized emission slightly increases while fluorine concentration grows. The presence of mixed anions, fluorine and oxygen, concerns at once the coordination-shell erbium ions and the host tellurite glass built of

three different structural units for the former network and has an influence by increasing the site-to-site change and modifying the ion–host field strengths. The particular feature of the fluorine richest tellurite glass compare to the other oxyfluoride should be explained by a different structural organization of the former network surrounding the Er^{3+} mainly constituted of entities more close as those found in ZnTeO_3 or $\text{Zn}_2\text{Te}_3\text{O}_8$. Additionally, the Er^{3+} lifetime of the emitting level $^4\text{I}_{13/2}$ increases drastically with fluorine content due mainly to the radiative rate change and by diminishing non-radiative decay rates related to the water vibrational frequency. As a consequence of the energy gap between $^4\text{I}_{11/2}$ and $^4\text{I}_{13/2}$ levels, the lifetime of the $^4\text{I}_{11/2}$ level is strongly influenced by the phonon energy. Due to lower probability of non-radiative multiphonon relaxation to $^4\text{I}_{13/2}$ level, this behavior more marked as the fluorine content increases could reduce the efficiency of an amplifier pumping at 980 nm. Introducing new elements such as GeO_2 which give an highest phonon energy or co-doping of cerium ion into the Er-doped low-phonon energy glasses should improved the 980 nm excitation efficiency by depopulating the $^4\text{I}_{11/2}$ level and will be the next focus of investigation in oxyfluoride tellurite glasses [7–9].

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